

Small, Bias-Free, Blind and Convolutional Denoiser

A compact ConvNeXt U-Net for blind Gaussian color-image denoising

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Abstract

We describe and evaluate BF-CONVUNEXT, a compact bias-free CONVNEXT U-Net for blind additive-white-Gaussian-noise (AWGN) color image denoising. It combines four ingredients, none of them new, so that a single property survives end to end. A frozen depthwise Gabor stem gives an oriented band-pass front end with zero trainable parameters. A Laplacian-pyramid encoder routes the full-resolution high-frequency residual into each skip connection, with three extra CONVNEXT blocks per band. A CONVNEXT-V1 U-Net forms the body. The construction is bias-free throughout: no additive bias in any convolution, a linear head, LeakyReLU activations, and a variance-only batch norm. Together these make the 0.82M-parameter network exactly degree-1 homogeneous at inference, $D(\alpha y) = \alpha D(y)$, which licenses the Miyasawa/Tweedie reading of the residual as a score and, following Mohan et al., supports blind generalization across noise levels from one model. We train a single blind model on a noise- σ curriculum spanning $\sigma_{255} \approx 6.4$ to 64; it extrapolates far past that ceiling without a cliff, degrading smoothly to 22.8 dB at $\sigma_{255}=150$ and 20.0 dB at $\sigma_{255}=200$ ($2.3\times$ and $3.1\times$ the training maximum). Evaluated unchanged on the four standard color sets (CBSD68, Kodak24, McMaster, Urban100) at $\sigma \in \{15, 25, 50\}$ under the full-image protocol, it matches or beats DnCNN and FFDNet on every set and level, averaging about +0.7 dB over DnCNN while FFDNet, though non-blind, still trails. The closest case is CBSD68: there the +0.10 to +0.15 dB margin over DnCNN falls within the sensitivity of our disclosed $[0, 1]$ input clip, so the result is best read as a match (Section 6). Against the heavyweight CNN/transformer state of the art it sits a small, consistent margin behind (~ 0.3 – 0.5 dB on CBSD68, ~ 0.3 – 0.7 dB on Kodak24, ~ 0.5 – 0.9 dB on McMaster, and ~ 0.6 – 1.7 dB on Urban100), at roughly 1/15 to 1/39 of their parameters. The homogeneity is inference-only and checkpoint-specific, and the learned residual is a *local* score, not a global prior (its Jacobian is non-conservative), so plug-and-play/RED and calibrated-posterior guarantees do not transfer. That same local score still drives stochastic sampling from the implicit prior and a range of linear inverse problems (inpainting, super-resolution, deblurring, compressive sensing). Results come from a single checkpoint, reported in PSNR (plus SSIM for our own model), without ablations.

1 Introduction

Blind Gaussian image denoising recovers a clean image x from $y = x + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$, with σ unknown at test time. It is both a canonical low-level vision task and a common building block inside modern inverse-problem solvers. This paper is framed by two facts. The field’s accuracy is dominated by large transformer/CNN hybrids (DRUNet, SwinIR, Restormer, SCUNet) with tens of millions of parameters, often trained per-noise-level. Set against them is a much older and simpler observation, still underexploited in a modern architecture: removing all additive constants from a denoising CNN makes it exactly scale-equivariant, and therefore able to generalize blindly across noise levels that a biased network cannot handle [Mohan et al., 2020].

This paper takes the second fact seriously and asks one empirical question: how far does a compact, strictly bias-free, modern-conv-block U-Net get on the standard color-denoising leaderboard as a single blind model? Our answer is BF-CONVUNEXT, a 0.82M-parameter bias-free CONVNEXT U-Net that composes four ideas we did not invent:

- a frozen depthwise Gabor stem [Gabor, 1946]: a bank of oriented, frequency-tuned band-pass filters fixed at initialization (zero trainable parameters), whose outputs are remixed into the stem channels by a single trainable bias-free 1×1 projection;

- a Laplacian-pyramid encoder [Burt & Adelson, 1983] that, at each downsampling junction, sends the full-resolution high-frequency residual band to the skip connection and passes the low-frequency band downward, with three extra CONVNEXT blocks applied to each high-frequency skip band;
- a CONVNEXT-V1 U-Net body [Liu et al., 2022, Ronneberger et al., 2015] (depthwise 7×7 conv, inverted-bottleneck MLP, LayerScale);
- bias-freedom from end to end: no additive bias in any layer, a linear head, LeakyReLU activations, and a variance-only batch normalization, chosen together so the network is globally degree-1 homogeneous at inference.

The contribution is the specific, bias-preserving way they are combined so that homogeneity survives end to end, together with an apples-to-apples measurement against the published leaderboard on the same test sets and full-image protocol. This is an integration-and-evaluation paper, closer to an engineering report than to a novel-component paper; it makes three contributions:

1. a precise description of BF-CONVUNEXT (topology, frozen Gabor stem, Laplacian-pyramid skip decomposition, CONVNEXT-V1 block, and the bias-free construction that makes the network degree-1 homogeneous), at a level of detail sufficient to reimplement it;
2. a fair benchmark: one blind checkpoint, evaluated unchanged on all four standard color sets at three noise levels under the full-image protocol, tabulated against six published baselines;
3. an explicit account of the theory’s limits: the Miyasawa/Tweedie residual-as-score reading is only a motivation, and we show (Section 3) that the learned operator is non-conservative, so it is not the gradient of any global prior.

The bias-free construction also reaches beyond denoising. Because the residual is a Miyasawa score, the same frozen model serves as the implicit prior in the stochastic inverse-problem solver of Kadkhodaie & Simoncelli [2021], extending with no retraining to linear inverse problems and to sampling from the implicit prior (Section 6). This is not an ablation study, and the leaderboard comparison is PSNR-only because the collated source is (we add SSIM for BF-CONVUNEXT in Table 2). The result is favorable: the compact blind model matches or beats the classic sub-1M CNNs (a clear win on Kodak24, McMaster, and Urban100; a match on CBSD68, Section 6) and lands within about a dB of models an order of magnitude larger.

2 Architecture

BF-CONVUNEXT is a two-level U-Net operating on images normalized to $[0, 1]$. All parameter counts and layer shapes below are read from the actual trained checkpoint (20260720_convunext_denoiser_hfb3, 821,832 parameters total: 807,048 trainable and 14,784 frozen). Figure 1 shows the data flow.

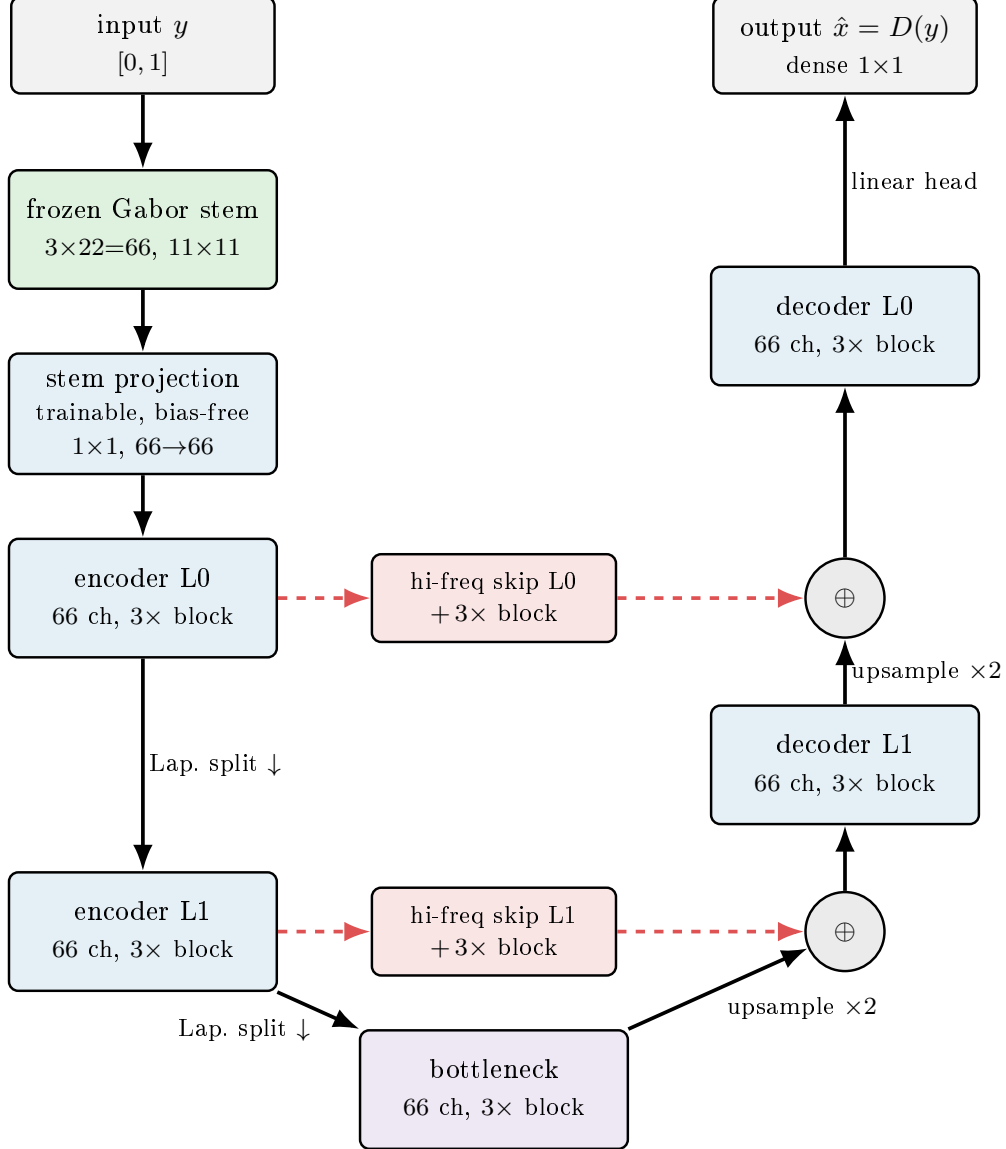


Figure 1: BF-CONVUNEXT data flow (channel widths shown). The encoder (left) descends, the decoder spine (right) ascends, and the bottleneck sits at the base. A frozen Gabor stem, followed by a trainable bias-free 1×1 projection (`gabor_stem_projection`) that remixes its 66 responses, feeds a two-level CONVNEXT-V1 U-Net. Downsampling is a Laplacian-pyramid split: the low-frequency band descends (solid arrows) while the full-resolution high-frequency residual becomes the skip (dashed), passing through three extra bias-free CONVNEXT blocks. At each decoder level the skip and the upsampled feature are combined at an explicit parameter-free addition ($\oplus$): the upsampled feature is added directly to the skip (both carry 66 channels, so their widths already agree), and only then is the sum processed by the decoder block. The head is a bias-free dense 1×1 convolution. Every operation is degree-1 homogeneous at inference, so $D(\alpha y) = \alpha D(y)$.

2.1 U-Net topology

Every scope runs at a flat width of 66 channels: the width multiplier is 1, so there is no doubling between levels and the two encoder levels, the bottleneck, and the two decoder levels all operate at 66 channels. Each of the five scopes (two encoder levels, bottleneck, two decoder levels) contains three CONVNEXT-V1 blocks. Upsampling is bilinear ($\times 2$). Decoder skips are fused by addition, $\text{dec} \leftarrow \text{Add}(\text{skip}, \text{up})$, not by concatenation; because every scope runs at the same 66 channels the skip and upsampled feature already agree in width, so no channel projection is needed. Both choices keep the merge exactly linear and homogeneous.

2.2 Frozen Gabor stem

In a plain CNN the first learnable operation on a noisy image is a small learned convolution. Here it is replaced by a fixed oriented band-pass front end. A depthwise convolution with 22 filters per input channel and an 11×11 kernel is initialized with a Gabor bank, each output channel being one 2D Gabor wavelet $g(u, v) = \exp\left(-\frac{u^2 + \gamma^2 v^2}{2s^2}\right) \cos\left(\frac{2\pi u \theta}{\lambda} + \psi\right)$ at a swept orientation, frequency, and phase, and is marked non-trainable, so it contributes no trainable parameters and never adapts. With three input channels it emits $3 \times 22 = 66$ feature maps. The rationale is standard signal processing: an oriented multi-scale band-pass decomposition is a strong, noise-agnostic image representation that the network need not spend capacity (or risk overfitting noise) to relearn.

Those 66 frozen responses are not handed to the encoder as they stand. They first pass through `gabor_stem_projection`, a single trainable bias-free 1×1 convolution ($66 \rightarrow 66$ channels, `use_bias=False`, $66 \times 66 = 4,356$ parameters), whose output is what we call the 66 stem channels. The division of labor is deliberate: the bank itself is a fixed, hand-designed dictionary that cannot adapt, while the projection is the one learned degree of freedom on top of it, free to reweight and combine band-pass responses across orientations, frequencies, phases, and the three input color channels without altering a single filter. Being a bias-free linear map it is degree-1 homogeneous and so leaves the property of Section 2.5 intact. It is cheap in parameters and less cheap in compute: 4,356 parameters is 0.5% of the model, but because a dense 1×1 convolution runs at full input resolution it costs ~ 0.57 GFLOP at 256^2 , about 1.0% of the model’s total forward FLOPs (Section 6).

2.3 Laplacian-pyramid skips and high-frequency blocks

Instead of max-pooling, each encoder junction performs a Laplacian-pyramid split. Given the level input z , a fixed Gaussian blur (5×5) and stride-2 blur-pool produce the low band $\ell = \text{blurpool}(\text{gauss}(z))$, and the high band is the full-resolution residual $h = z - \text{up}(\ell)$. The split is lossless ($\text{up}(\ell) + h = z$) and channel-preserving. The low band ℓ descends to the next level. The high band h becomes the skip connection, but only after passing through three additional bias-free CONVNEXT-V1 blocks (`skip_highfreq`), applied at both encoder levels. This is a wavelet-shrinkage-style inductive bias. Gaussian noise is spectrally flat while natural-image structure concentrates at low frequencies, so forcing the decoder to reconstruct from an explicit low-band descent plus a processed high-band residual denies it the trivial full-resolution identity-copy shortcut and matches the signal/noise spectral structure of the task.

2.4 The CONVNEXT-V1 block

Each block is a residual branch with call order

$$x \mapsto x + \gamma \odot \left[W_{\downarrow} \phi \left(W_{\uparrow} \text{N}(\text{DW}_{7 \times 7} x) \right) \right],$$

where $\text{DW}_{7 \times 7}$ is a depthwise 7×7 convolution, N is the normalization (Section 2.5), $W_{\uparrow}$ is a 1×1 expansion to $4 \times$ width, ϕ is LeakyReLU with negative slope 0.1 (followed by dropout 0.1 at training time), $W_{\downarrow}$ is a 1×1 reduction, and γ is a per-channel LayerScale multiplier (initialized 10^{-4} , floored at 10^{-6}). All convolutions are bias-free. We use the V1 block rather than V2, because the V2 variant inserts a Global Response Normalization whose learnable additive β is bias-like and would break strict bias-freedom.

2.5 Bias-free construction and degree-1 homogeneity

The defining property of BF-CONVUNEXT is that the entire map D is positively homogeneous of degree one at inference:

$$D(\alpha y) = \alpha D(y) \quad \text{for all } \alpha > 0. \quad (1)$$

This holds because every constituent operation is linear-through-the-origin or degree-1 homogeneous, and no operation injects an additive constant:

- every convolution (Gabor stem, 1×1 mix, all block convolutions, the head) has `use_bias=False`;
- the head activation is linear ($f(x) = x$);
- LeakyReLU is positively homogeneous, $\text{LReLU}(\alpha x) = \alpha \text{LReLU}(x)$ for $\alpha > 0$ (this is why LeakyReLU is used rather than GELU, which is not homogeneous);
- normalization is a *variance-only* batch norm, `BIASFREEBATCHNORM`: it has no running mean and no β offset, only a fixed non-trainable running variance, so at inference $N(x) = \gamma x / \sqrt{\text{Var} + \epsilon}$ is exactly linear in x ;
- the Laplacian split, bilinear upsampling, and additive skip merge are all linear.

The choice of `BIASFREEBATCHNORM` over the `CONVNEXT` default `LayerNorm` is not cosmetic. `LayerNorm` divides by a per-sample statistic that itself scales with the input, making it scale-invariant (degree-0), which would destroy Eq. (1) for the whole network. Homogeneity is an inference-time property: during training `BIASFREEBATCHNORM` uses the per-batch variance and is therefore degree-0, so Eq. (1) holds only once the running statistics are frozen.

3 Why bias-free: Miyasawa/Tweedie, and its limits

Bias-freedom is more than an aesthetic choice: it connects the trained denoiser to a score, and it lets one blind model span a wide noise range. We give the motivation first, then its boundaries.

Miyasawa/Tweedie. For $y = x + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2 I)$ and marginal density $p_\sigma(y) = \int \mathcal{N}(y; x, \sigma^2 I) p(x) dx$, the minimum-MSE estimator obeys the Miyasawa/Tweedie identity

$$\hat{x}(y) = \mathbb{E}[x | y] = y + \sigma^2 \nabla_y \log p_\sigma(y). \quad (2)$$

Hence for an MMSE-optimal D the denoiser residual $f(y) = D(y) - y$ is exactly σ^2 times the score $\nabla_y \log p_\sigma(y)$ of the noise-blurred density. A plain mean-squared-error objective is therefore well motivated: it drives D toward the MMSE estimator whose residual is a scaled score. This is why BF-CONVUNEXT is trained with MSE and read as an implicit-prior score model, with the caveat (made precise below) that the *learned* residual realizes this only as a *local* score estimate, not the gradient of a global density.

Homogeneity buys blind generalization. Mohan et al. [2020] observe that a bias-free denoiser is degree-1 homogeneous (Eq. (1)) and therefore behaves as a locally linear, input-adaptive filter $D(y) \approx J(y) y$ whose Jacobian $J(y) = \partial D / \partial y$ is degree-0 (scale-invariant). Because J depends only on the direction of y , not its magnitude, a network trained over one range of noise levels extrapolates gracefully to unseen levels. A biased network, whose additive terms fix an absolute scale, instead collapses off its training range. This is the property that makes a single blind model viable, and it is the empirical target of our noise curriculum (Section 4).

The limits of the homogeneity argument. Eq. (2) is easy to over-read, in two ways.

1. **Homogeneity is checkpoint-specific, not an architectural guarantee.** A network is exactly homogeneous only if every operation is genuinely bias-free. For this checkpoint the relative homogeneity error $\|D(\alpha y) - \alpha D(y)\|/\|\alpha D(y)\|$ is exactly 0 across $\alpha \in \{\frac{1}{4}, \frac{1}{2}, 2, 4, 8\}$ (these are powers of two, for which the scaling is exact in binary floating point), and it stays at the level of floating-point round-off, below 1×10^{-4} , under a non-dyadic stress test at $\alpha \in \{0.3, 1.3, 3.7, 6.1\}$. The property is fragile: a single non-bias-free operation, such as a LayerNorm, drives the homogeneity error up by four orders of magnitude, to order ~ 0.9 . Eq. (1) must therefore be verified per checkpoint, not assumed.
2. **The learned residual is not a conservative field.** Eq. (2) would make $f(y)$ the gradient of the scalar $\sigma^2 \log p_\sigma(y)$ only if the Jacobian $J(y)$ were symmetric, and it is not. For this checkpoint the measured Jacobian asymmetry $\|J - J^\top\|/\|J\|$ is 0.93, roughly three orders of magnitude ($\sim 2400\times$) above a symmetric-blur control, so no global energy or log-density function exists whose gradient is the residual. Plug-and-play and Regularization-by-Denoising convergence guarantees assume a conservative, symmetric-Jacobian denoiser, so they do not transfer here, and the residual should be read as a useful local score estimate rather than a calibrated global prior.

We return to the practical consequences in Section 6.

4 Training

Data and noise curriculum. We train on 256×256 patches drawn from COCO (`train2017`) and DIV2K (`train`), sourced with a per-directory weighting so the $\sim 118\text{K}$ COCO images do not drown DIV2K’s ~ 800 . Patches are normalized to $[0, 1]$ (peak-to-peak range 1.0) and corrupted with additive Gaussian noise whose per-patch standard deviation is drawn uniformly from $[0, \sigma_{\max}]$, where the upper bound $\sigma_{\max}$ is widened linearly over training by a curriculum callback from 0.025 to 0.25 across 100 epochs. Starting narrow and widening stabilizes early optimization, and bias-freedom then supplies the cross-level generalization. The noisy input is clipped to $[0, 1]$ immediately after the noise is added, so no pixel is ever presented to the network above 1 or below 0; the identical clip is applied at inference, so the model operates on the same bounded domain in both phases. (At large σ this makes the effective corruption a clipped Gaussian rather than a purely additive one.)

The σ convention. The model operates entirely in $[0, 1]$, so its noise level $\sigma_{\text{norm}} \in [0, 0.25]$. The literature reports noise on the $[0, 255]$ pixel scale, and the two are related by $\sigma_{255} = \sigma_{\text{norm}} \cdot 255$. The curriculum’s $\sigma_{\max} : 0.025 \rightarrow 0.25$ therefore spans $\sigma_{255} \approx 6.4 \rightarrow 64$, which covers the classic 15/25/50 benchmark regimes as a single blind model. All PSNR values are computed with `max_val = 1.0` (matched to the peak-to-peak range). Because PSNR is scale-invariant under a matched range, these dB numbers are directly comparable to published `max_val = 255` results, with no further conversion.

Optimization. Loss is plain MSE (the Miyasawa-optimal objective, Section 3). We use AdamW (learning rate 10^{-3} , cosine decay, 10 warmup epochs, decoupled weight decay 4×10^{-3}), gradient-norm clipping at 1.0, batch size 4, 100 epochs.

5 Results

We report two evaluations of the single trained checkpoint. (i) A blind PSNR-vs-noise sweep on DIV2K-validation (100 patches at 256px). (ii) The standard leaderboard comparison on all four color sets (CBSD68, Kodak24, McMaster, Urban100) at $\sigma \in \{15, 25, 50\}$ using the full-image protocol (whole images, reflect-padded to a multiple of 16), the same protocol the published numbers use. One caveat applies: our inputs are clipped to $[0, 1]$ (Section 4), whereas the collated baselines follow the more common unclipped-AWGN convention, and we quantify the effect of this asymmetry in Section 6.

5.1 Blind denoising across noise levels

Figure 2 shows the blind PSNR-vs-noise sweep on DIV2K-validation, where bias-freedom matters most. A single model, given no noise-level input, denoises across the whole curriculum range (40.9dB at $\sigma_{255}=5$ to 28.9dB at $\sigma_{255}=65$, just past the $\sigma_{255}\approx 64$ training maximum), and it keeps working *far beyond* that maximum. The curriculum only ever showed the network $\sigma_{\text{norm}} \leq 0.25$; evaluated at $\sigma_{\text{norm}}=0.59$ and 0.78 ($\sigma_{255}=150$ and 200 , $2.3\times$ and $3.1\times$ the largest training noise) it still returns 22.8 and 20.0 dB, degrading smoothly and monotonically with no cliff even at more than three times the training ceiling. This is the extrapolation predicted by the degree-1 homogeneity of Section 3, which Mohan et al. [2020] argue a *biased* network cannot manage. (Gains are quoted against the noisy input *as fed to the model*, clipped to the $[0, 1]$ domain, which at high σ has a higher PSNR than the unclipped Gaussian.) Figure 3 plots the same sweep against the noisy-input baseline, making the per-level denoising gain explicit.

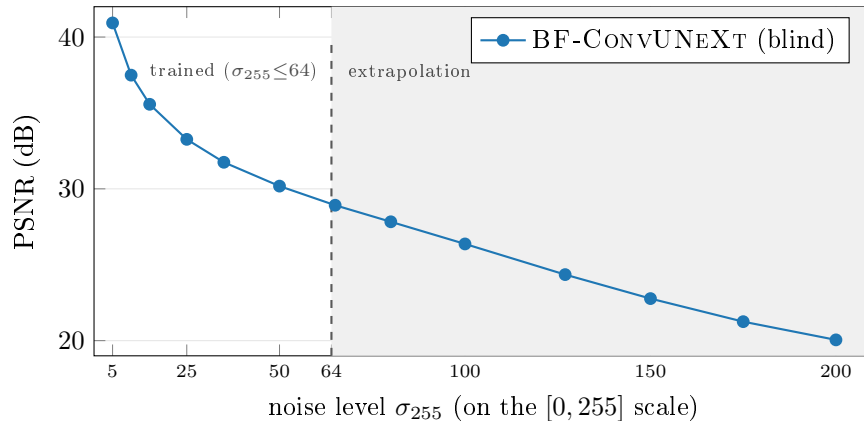


Figure 2: Blind PSNR vs. noise on DIV2K-validation (100 patches, 256px). One model, no noise-level input, across the full range and **beyond it**: the curriculum trained only up to $\sigma_{255}\approx 64$ (dashed line), yet the model keeps denoising in the shaded extrapolation region: 22.8dB at $\sigma_{255}=150$ and 20.0dB at $\sigma_{255}=200$ ($2.3\times$ and $3.1\times$ the training maximum), degrading smoothly with no cliff. This graceful off-range behaviour is the signature of the bias-free/degree-1-homogeneous construction (Section 3); a biased network collapses once past its training noise levels. The 95% bootstrap CIs (not shown) are ± 0.4 – 0.8 dB.

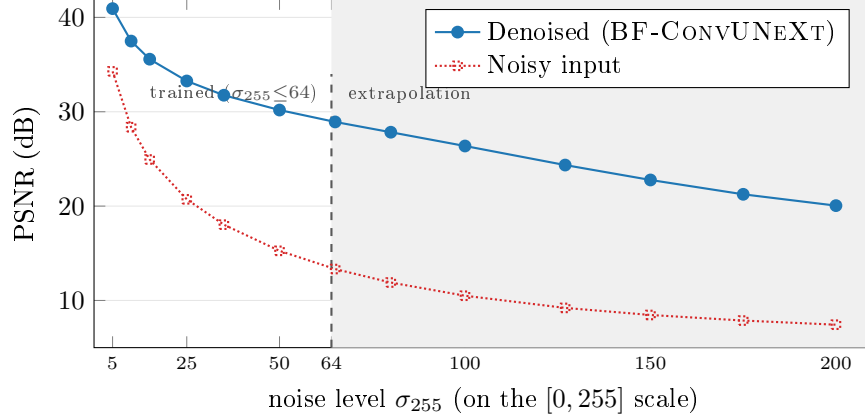


Figure 3: Denoising gain: denoised PSNR (solid blue) vs. the corrupted noisy-input PSNR (dotted red) across the noise sweep on DIV2K-validation (100 patches, 256px). The vertical gap between the two curves is the per-level PSNR gain from a single blind model: ~ 6.7 dB at $\sigma_{255}=5$, widening to ~ 15.6 – 15.9 dB around $\sigma_{255}=65$ – 80 . The gain persists far past the $\sigma_{255}\approx 64$ training ceiling (dashed line; shaded extrapolation region), still $+12.6$ dB at $\sigma_{255}=200$ (20.0 vs. 7.4 dB), consistent with the graceful off-range behaviour in Figure 2.

5.2 Comparison against the color-denoising leaderboard

Table 1 places BF-CONVUNEXT against six published baselines on all four standard color sets at $\sigma \in \{15, 25, 50\}$, under the identical full-image protocol and PSNR convention. The single 0.82M blind model matches or beats DnCNN and FFDNet on every set at every noise level, and trails the heavyweight state of the art by a small, consistent margin that widens on Urban100.

Table 1: Color AWGN denoising, PSNR (dB), full-image protocol, at $\sigma \in \{15, 25, 50\}$. Baseline numbers are the published values collated by Zhang et al. [2022]. **Bold** marks where BF-CONVUNEXT beats a given baseline. Approximate parameter counts are listed in the header; DnCNN/FFDNet are < 1 M, the four rightmost are 12–32M.

Set	σ_{255}	BF-CONVUNEXT (0.82M)	DnCNN (0.6M)	FFDNet (0.5M)	DRUNet (32M)	SwinIR (12M)	Restormer (26M)	SCUNet (18M)
CBSD68	15	34.00	33.90	33.87	34.30	34.42	34.40	34.40
CBSD68	25	31.35	31.24	31.21	31.69	31.78	31.79	31.79
CBSD68	50	28.10	27.95	27.96	28.51	28.56	28.60	28.61
Kodak24	15	34.99	34.60	34.63	35.31	35.34	35.47	35.34
Kodak24	25	32.51	32.14	32.13	32.89	32.89	33.04	32.92
Kodak24	50	29.35	28.95	28.98	29.86	29.79	30.01	29.87
McMaster	15	34.86	33.45	34.66	35.40	35.61	35.61	35.60
McMaster	25	32.59	31.52	32.35	33.14	33.20	33.34	33.34
McMaster	50	29.44	28.62	29.18	30.08	30.22	30.30	30.29
Urban100	15	34.24	32.98	33.83	34.81	35.13	35.13	35.18
Urban100	25	31.85	30.81	31.40	32.60	32.90	32.96	33.03
Urban100	50	28.50	27.59	28.05	29.61	29.82	30.02	30.14
avg $\sigma=15$		34.52	33.73	34.25	34.96	35.13	35.15	35.13
avg $\sigma=25$		32.07	31.43	31.77	32.58	32.69	32.78	32.77
avg $\sigma=50$		28.85	28.28	28.54	29.52	29.60	29.73	29.73

Averaged over the four sets, BF-CONVUNEXT is $+0.79/+0.64/+0.57$ dB above DnCNN at $\sigma = 15/25/50$

and $+0.27/+0.30/+0.31$ dB above FFDNet (which, unlike our model, is *non-blind* and receives a noise-level map). It is $-0.44/-0.51/-0.67$ dB relative to DRUNet and $-0.63/-0.71/-0.88$ dB relative to Restormer, the leaderboard top. BF-CONVUNEXT is ahead of DnCNN in all twelve cells, but the per-set margin is uneven: it is narrow on CBSD68 ($+0.10$ to $+0.15$ dB) and much larger on McMaster and Urban100 (up to $+1.41$ and $+1.26$ dB at $\sigma=15$). On CBSD68 that margin is within the sensitivity of our $[0, 1]$ input clip, so the result there is best read as a match rather than a clear win (Section 6); the McMaster, Urban100, and Kodak24 margins are far larger. Against the heavyweight state of the art the gap is smallest on CBSD68 (~ 0.3 – 0.5 dB), then Kodak24 (~ 0.3 – 0.7 dB) and McMaster (~ 0.5 – 0.9 dB), and largest on Urban100 (~ 0.6 – 1.7 dB). Urban100’s repetitive, self-similar structures are exactly where the transformers’ non-local attention (SwinIR, Restormer) and DRUNet’s large receptive field have the most to exploit and a compact local conv U-Net the least.

5.3 Structural similarity (SSIM)

Table 2 reports SSIM alongside PSNR for BF-CONVUNEXT on the same four sets and full-image protocol (same checkpoint as Table 1), computed on the $[0, 1]$ representation with a data range of 1.0. SSIM tracks PSNR’s ordering across noise, highest at $\sigma_{255}=15$ (average 0.9318) and degrading gracefully to average 0.8221 at $\sigma_{255}=50$. Urban100 is a cross-metric exception: it has the highest SSIM of the four sets at every noise level (0.945/0.918/0.857) despite its middling PSNR. Its regular, high-contrast man-made structures, the same self-similarity that costs PSNR against non-local models, yield strong local structural agreement once denoised. We report SSIM for BF-CONVUNEXT only, because our comparison source [Zhang et al., 2022] tabulates PSNR alone and SSIM values assembled from mixed sources use inconsistent windowing, border, and color conventions, so a baseline SSIM column would be an unreliable comparison.

Table 2: BF-CONVUNEXT PSNR (dB) and SSIM on the four color benchmark sets, full-image protocol, single blind checkpoint (identical protocol and model as Table 1; the PSNR columns reproduce it exactly). SSIM on $[0, 1]$, data range 1.0. Baseline SSIM is not tabulated (see Section 5.3).

Set	$\sigma_{255} = 15$		$\sigma_{255} = 25$		$\sigma_{255} = 50$	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
CBSD68	34.00	0.9316	31.35	0.8870	28.10	0.7981
Kodak24	34.99	0.9259	32.51	0.8855	29.35	0.8061
McMaster	34.86	0.9241	32.59	0.8918	29.44	0.8269
Urban100	34.24	0.9454	31.85	0.9178	28.50	0.8572
<i>average</i>	34.52	0.9318	32.07	0.8955	28.85	0.8221

5.4 Qualitative examples

Figure 4 shows the model on three held-out benchmark images (a Kodak24 scene, an Urban100 facade with fine repetitive structure that is a known denoiser stress case, and a McMaster crop with saturated color) across a wide sweep of noise, $\sigma_{255} \in \{50, 100, 200\}$, from moderate corruption to near-total static. The highest level, $\sigma_{255}=200$, is $3.1\times$ the training ceiling ($\sigma_{255}\leq 64$); the same frozen blind model is applied at every level with no noise-level input. Full-image inference is run under the paper’s protocol; the panels are a detail crop of the result. Even where the input is almost pure noise, and outside the trained range, edges and repeated structure come back cleanly, matching the out-of-range curve in Figure 2. Per-image PSNR/SSIM (computed on the full image) are annotated.

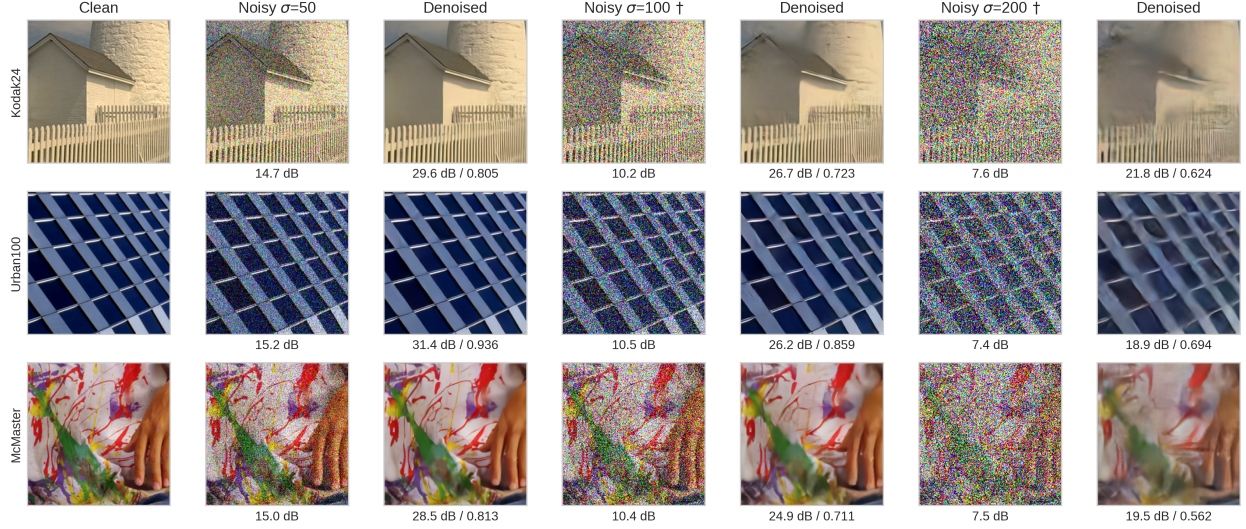


Figure 4: A clean reference followed by (noisy, denoised) pairs at $\sigma_{255} \in \{50, 100, 200\}$ on three held-out benchmark images (rows: Kodak24, Urban100, McMaster). One frozen blind model, no noise-level input, at every level. † marks $\sigma_{255} \in \{100, 200\}$, beyond the $\sigma_{255} \leq 64$ training range: from about 7.5dB of static at $\sigma_{255}=200$ the model still returns a 19–22 dB image. Per-image PSNR and SSIM (full image, max_val=1) are annotated under each panel.

5.5 Accuracy for the parameter budget

The point of Table 1 is not that BF-CONVUNEXT wins outright, because it does not, but where it sits on the accuracy/size trade-off. Figure 5 plots the four-set average PSNR at $\sigma = 25$ against approximate parameter count. BF-CONVUNEXT lands between the two size classes: comfortably above the sub-1M classic CNNs it beats, and within about a dB of models 15–39× larger.

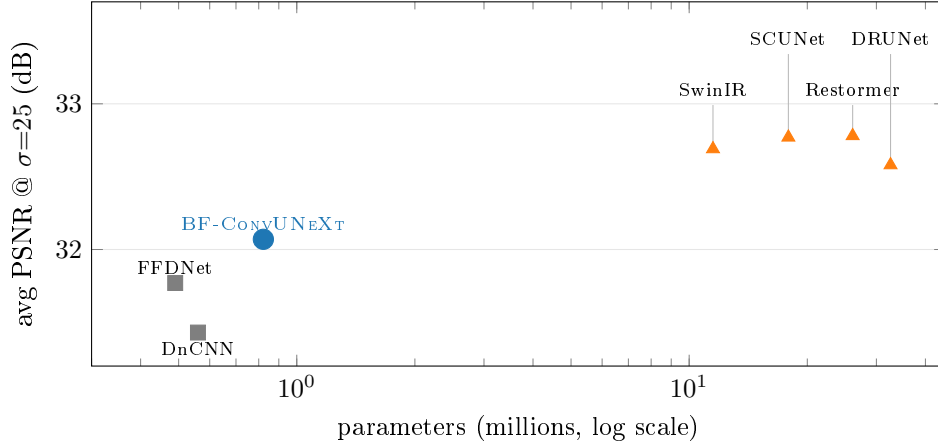


Figure 5: Four-set average PSNR at $\sigma=25$ vs. (approximate) parameter count. BF-CONVUNEXT (blue) sits on the efficient side of the trade-off: above the sub-1M CNNs and within ~ 0.71 dB of the 12–32M state of the art at $\sim 1/15$ – $1/39$ of their size. Parameter counts are approximate and collated from the respective papers.

The parameter count is only half the efficiency story; Table 3 adds compute and memory, measured on one RTX 4090. At 256^2 the model is ~ 59 GFLOPs and peaks at ~ 582 MiB of GPU memory; FLOPs and memory both scale with pixel count (a fully-convolutional model), so a 512^2 image is exactly 4× the

FLOPs and $\sim 4.0\times$ the peak memory. Wall-clock latency, however, does not track pixel count at batch 1: a single-image forward pass takes ~ 0.12 s at both resolutions (118 and 118ms median, indistinguishable despite $4\times$ the compute). At batch 1 the forward pass is dominated by kernel-dispatch and launch overhead rather than by compute, so it sits on a latency floor, and batched throughput is substantially higher. The “compact” claim is about parameters, FLOPs, and memory, and does not lean on any low-precision trick.

Table 3: Efficiency of BF-CONVUNEXT, measured on one RTX 4090 (single-image forward pass). FLOPs are per forward pass; memory is the peak GPU allocation. Cost is linear in pixel count. Latency is not tabulated because at batch 1 it sits on a dispatch-overhead floor rather than tracking compute (see text).

Resolution	Parameters	FLOPs	FLOPs/megapixel	Peak GPU mem
256×256	0.82M	59.26 GFLOP	904.18 GFLOP	581.7 MiB
512×512	0.82M	237.02 GFLOP	904.18 GFLOP	2312.2 MiB

6 Discussion: limitations and what does not work

Where the model stands. As a compact, blind denoiser BF-CONVUNEXT is strong: it matches or beats the classic sub-1M CNNs everywhere (a clear win on three of the four sets; a match on CBS68 within the input-clip protocol margin) and closes most of the gap to models an order of magnitude larger, from a single checkpoint with no noise-level input. It does not beat the heavyweight state of the art, and the gap is structurally largest on Urban100, where non-local self-similarity rewards the attention and large receptive fields that a two-level local conv U-Net lacks.

Why parameter count understates cost. Two of BF-CONVUNEXT’s layers illustrate why a “compact” claim should not rest on parameters alone. The trainable stem projection (Section 2) and the dense 1×1 head are both 1×1 convolutions evaluated at *full input resolution*, where a one-pixel kernel still costs one multiply-accumulate per output pixel per channel pair. Together they are 4,554 parameters, 0.6% of the model, but ~ 0.6 GFLOP at 256^2 , or $\sim 1.0\%$ of the model’s total forward FLOPs — roughly a two-fold gap between the two framings of the same two layers, and the gap widens with resolution while the parameter share stays fixed. We therefore report FLOPs and peak memory alongside parameters in Table 3, and read the parameter column as the least informative of the three. The latency column is omitted deliberately: at batch 1 the forward pass is dispatch-bound, so the 512^2 median is not larger than the 256^2 median despite $4\times$ the compute, and quoting it would misrepresent the cost model.

Limits of the theory. The one genuinely negative result is the theory limit of Section 3: the learned residual is a local score, not the gradient of a global density, so uses that require a conservative denoiser (provable plug-and-play/RED convergence, energy-based sampling toward a fixed target, a calibrated Bayesian posterior) are out of reach. The homogeneity that does hold is inference-only and must be re-verified per checkpoint.

What the local score still buys. That the residual is only a local score, not a global prior, is less limiting than it may appear. In the annealed coarse-to-fine stochastic ascent of Kadkhodaie & Simoncelli [2021], the denoiser residual is used directly as the score at each effective noise level ($y \leftarrow y + h_t d_t + \gamma_t z_t$ with $d_t = (I - MM^\top) f(y) + M(m - M^\top y)$ for a linear measurement operator M with orthonormal columns), and this loop nowhere assumes a symmetric Jacobian or a global energy. The same frozen model, unchanged, therefore drives unconstrained sampling from the implicit prior (the $M=0$ case) and solves a family of linear inverse problems (inpainting, random-pixel recovery, super-resolution, spectral deblurring, compressive sensing) from one blind denoiser. Unlike plug-and-play/RED it offers no fixed-point convergence guarantee and needs no conservative operator; it is an empirical sampler. A local score, though weaker than a global prior, therefore still supports these downstream uses.

Measurement limits. We report PSNR and SSIM for BF-CONVUNEXT (Tables 1 and 2), but the baseline comparison is PSNR-only, because our collated source [Zhang et al., 2022] reports PSNR and cross-source SSIM is protocol-inconsistent. Table 1 shows point PSNR without per-cell error bars: with a fixed noise seed each cell is a deterministic function of the frozen model, so the relevant uncertainty is the across-image spread, which the DIV2K sweep reports as ± 0.4 – 0.8 dB 95% bootstrap CIs (Figure 2); these are indicative of, but not measured on, the leaderboard sets. All results are from a single checkpoint, and we present no ablations: the individual contributions of the frozen Gabor stem, the Laplacian high-frequency skips, the `high_freq_blocks`, the bias-free normalization, and the noise curriculum are not isolated. Our baselines are externally collated rather than run in-house; a bias-free flat CNN would supply the missing closest-competitor row and isolate the *architecture*, though not bias-freedom itself, for which the clean control is this same network trained *with* biases and a standard normalization. One protocol asymmetry deserves an explicit caveat: we clip the noisy input to $[0, 1]$ at both training and inference (Section 4), whereas the standard AWGN benchmark convention is unclipped additive noise, and the collated baselines almost certainly follow it. Because clipping removes the out-of-range excursions that carry pure error, the clipped input is an easier starting point. We measured the size of this advantage directly, by re-running the CBSD68 evaluation of this same checkpoint under the identical full-image protocol with the clip disabled: it is worth $+0.28$, $+0.71$, and $+2.23$ dB at $\sigma_{255}=15$, 25, and 50. This has two consequences. At every level the clip advantage exceeds our $+0.10$ to $+0.15$ dB margin over DnCNN on CBSD68, so we read that comparison as a match rather than a win. It also grows steeply with σ , since clipping discards a larger share of the noise as the noise grows, and by $\sigma_{255}=50$ it is several times the ~ 0.3 – 0.5 dB gap we report to the heavyweight state of the art on that set. The effect inflates only our own numbers, so it never narrows that gap; but it does mean that at the highest level our CBSD68 figures are not directly comparable to unclipped-protocol numbers, and a like-for-like unclipped comparison there would place us further behind the leaderboard than Table 1 suggests, not closer. We measured this on CBSD68 only. The McMaster, Urban100, and Kodak24 margins over DnCNN/FFDNet are wide enough to absorb an effect of the $\sigma_{255}=15$ – 25 magnitude, but we have not measured the clip advantage on those sets and do not claim their $\sigma_{255}=50$ margins survive it. A perceptual metric such as LPIPS, and grayscale Set12/BSD68 (which needs a one-channel model), are likewise outside the present scope. These are deliberate boundaries of a single-checkpoint study.

7 Related work

Bias-free denoising. Mohan et al. [2020] introduced bias-free CNNs for blind denoising and established the degree-1-homogeneity argument we build on. The residual-as-score reading rests on the classical Miyasawa/Tweedie identity [Miyasawa, 1961, Robbins, 1956], which also underpins the implicit-prior sampling line of Kadkhodaie & Simoncelli [2021]. BF-CONVUNEXT applies the bias-free discipline end-to-end to a modern block and front end.

CNN and transformer denoisers. DnCNN [Zhang et al., 2017] established the residual-CNN denoiser; FFDNet [Zhang et al., 2018] added a noise-level map for non-blind flexibility; DRUNet [Zhang et al., 2021] coupled a U-Net denoiser with plug-and-play restoration; SwinIR [Liang et al., 2021], Restormer [Zamir et al., 2022], and SCUNet [Zhang et al., 2022] brought (windowed/channel) attention and are the current leaderboard leaders. These are our baselines; all except our model and DnCNN’s blind variant either use a noise map or are larger by 15 – $39\times$.

Building blocks. The block is a CONVNEXT [Liu et al., 2022] in a U-Net [Ronneberger et al., 2015]. The multi-scale skip decomposition is a Laplacian pyramid [Burt & Adelson, 1983], and the frozen front end is a Gabor filter bank [Gabor, 1946]. BF-CONVUNEXT contributes their bias-preserving integration and its measurement.

8 Reproducibility

The entire stack is reproducible and open: the model, the training recipe, the noise curriculum, the full-image evaluation harness, and the implicit-prior inverse-problem solver of Section 6 are all implemented in

the `dl_techniques` framework at https://github.com/NikolasMarkou/dl_techniques.

The evaluated checkpoint is `20260720_convunext_denoiser_hfb3` (821,832 parameters total, 807,048 trainable and 14,784 frozen; depth 2; flat width 66; three CONVNEXT-V1 blocks per scope plus three per high-frequency skip band; frozen 22-filter, 11×11 Gabor stem followed by a trainable bias-free 1×1 stem projection; BIASFREEBATCHNORM; LeakyReLU(0.1); linear dense head). It was trained with the recipe of Section 4. The full-image benchmark uses whole test images reflect-padded to a multiple of 16, PSNR at `max_val` = 1.0 on $[0, 1]$, on the standard CBSD68 / Kodak24 / McMaster / Urban100 sets. All numbers in Table 1, Table 2 and Figure 2 are measured from that single checkpoint; baseline numbers are the published values as collated by Zhang et al. [2022]. The efficiency numbers (Table 3) are single-image forward passes on one RTX 4090.

Gabor stem, in full. The frozen stem is fully specified so it can be regenerated deterministically (given the grid and seed below; the exact per-axis `linspace` resolution and channel-ordering are fixed in the released code). It is a depthwise 11×11 convolution with depth multiplier 22 (so $3 \times 22 = 66$ output channels), `use_bias=False`, `trainable=False`. Each output channel is one real 2D Gabor wavelet $g(u, v) = \exp(-\frac{u_\theta^2 + \gamma^2 v_\theta^2}{2s^2}) \cos(\frac{2\pi u_\theta}{\lambda} + \psi)$, with (u_θ, v_θ) the coordinates rotated by θ . The five parameters are swept on a regular `linspace` grid over the Özbulak–Ekenel Table-I ranges ($s \in [2, 21]$, $\theta \in [0^\circ, 360^\circ]$, $\lambda \in [8, 100]$, $\gamma \in [0, 300\%]$, $\psi \in [0^\circ, 360^\circ]$), one distinct $(s, \theta, \lambda, \gamma, \psi)$ tuple per output channel, replicated identically across the three input channels, with global seed 42. Because the bank is fixed, it contributes zero trainable parameters and requires no checkpointing beyond these constants.

Stem projection, in full. The one trainable layer between the frozen bank and encoder L0 is `gabor_stem_projection`: a Conv2D with 66 filters, kernel 1×1 , stride 1, no padding needed, `use_bias=False`, `trainable=True`, linear activation, and no normalization. Its kernel is therefore a single 66×66 matrix (4,356 parameters), initialized with the framework default (Glorot uniform) under the same global seed 42, and it is trained jointly with the rest of the network under the recipe of Section 4. It applies the same 66×66 mix at every spatial position, so it is exactly a learned linear recombination of the frozen Gabor responses, and, being bias-free and linear, it preserves degree-1 homogeneity. The final head is a bias-free dense 1×1 Conv2D back to 3 channels (198 parameters, `groups = 1`, linear activation).

9 Conclusion

We described BF-CONVUNEXT, a compact bias-free CONVNEXT U-Net that integrates a frozen Gabor stem, Laplacian-pyramid high-frequency skips, and end-to-end degree-1 homogeneity, and we measured it against the color-denoising leaderboard. As a single 0.82M-parameter blind model it matches or beats the classic sub-1M CNNs on all four standard sets at all three noise levels (a match on CBSD68 once the input clip is accounted for, Section 6) and comes within about a dB of state-of-the-art models an order of magnitude larger, with the residual gap concentrated on self-similar imagery. Within its stated limits (a single, non-conservative checkpoint) it is a useful data point on how much denoising accuracy, and, through the implicit-prior score, how much downstream inverse-problem capability, is reachable from a small, strictly bias-free model.

References

- Burt, P. J., & Adelson, E. H. (1983). The Laplacian pyramid as a compact image code. *IEEE Transactions on Communications*, 31(4), 532–540.
- Gabor, D. (1946). Theory of communication. *Journal of the Institution of Electrical Engineers*, 93(26), 429–457.
- Kadkhodaie, Z., & Simoncelli, E. P. (2021). Stochastic solutions for linear inverse problems using the prior implicit in a denoiser. *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2007.13640.

- Liang, J., Cao, J., Sun, G., Zhang, K., Van Gool, L., & Timofte, R. (2021). SwinIR: Image restoration using Swin Transformer. *ICCV Workshops*. arXiv:2108.10257.
- Liu, Z., Mao, H., Wu, C.-Y., Feichtenhofer, C., Darrell, T., & Xie, S. (2022). A ConvNet for the 2020s. *CVPR*. arXiv:2201.03545.
- Miyasawa, K. (1961). An empirical Bayes estimator of the mean of a normal population. *Bulletin of the International Statistical Institute*, 38(4), 181–188.
- Mohan, S., Kadkhodaie, Z., Simoncelli, E. P., & Fernandez-Granda, C. (2020). Robust and interpretable blind image denoising via bias-free convolutional neural networks. *International Conference on Learning Representations (ICLR)*. arXiv:1906.05478.
- Robbins, H. (1956). An empirical Bayes approach to statistics. *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability*, 1, 157–163.
- Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional networks for biomedical image segmentation. *MICCAI*. arXiv:1505.04597.
- Zamir, S. W., Arora, A., Khan, S., Hayat, M., Khan, F. S., & Yang, M.-H. (2022). Restormer: Efficient transformer for high-resolution image restoration. *CVPR*. arXiv:2111.09881.
- Zhang, K., Zuo, W., Chen, Y., Meng, D., & Zhang, L. (2017). Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising. *IEEE Transactions on Image Processing*, 26(7), 3142–3155. arXiv:1608.03981.
- Zhang, K., Zuo, W., & Zhang, L. (2018). FFDNet: Toward a fast and flexible solution for CNN-based image denoising. *IEEE Transactions on Image Processing*, 27(9), 4608–4622. arXiv:1710.04026.
- Zhang, K., Li, Y., Zuo, W., Zhang, L., Van Gool, L., & Timofte, R. (2021). Plug-and-play image restoration with deep denoiser prior. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10), 6360–6376. arXiv:2008.13751.
- Zhang, K., Li, Y., Liang, J., Cao, J., Zhang, Y., Tang, H., Fan, D.-P., Timofte, R., & Van Gool, L. (2022). Practical blind image denoising via Swin-Conv-UNet and data synthesis. arXiv:2203.13278.